Part 20 — Multiphase and solid-state fermentation, gel immobilisation, the pathway-modification playbook, and design of experiments properly

I audited the module for the four things you named before answering. Two were well covered, one was thin, and two were absent — and the absent ones are whole reactor classes.

What you asked about	In the module	Verdict
Metabolic pathway modification	Knockouts and overexpression (158 mentions), dynamic control (82), compartmentalisation (75), cofactor swaps (57), feedback resistance (17)	**Well covered**, but scattered across twelve parts with no single decision guide — chapter V3 consolidates it
In situ gel fermentation	Part 3 chapter 48: alginate, Thiele modulus, dead core, effectiveness factor (60 mentions)	**Covered as transport**, not as a gel process — bead making, leakage and gel stability were missing (V2)
Two-liquid-phase and gas fermentation	Part 3 chapters 45–47, part 5 chapter 55 (54 and 16 mentions)	Covered
Three-phase (gas–liquid–solid) reactors	Nothing	**Absent** — V1
Solid-state fermentation	Nothing	**Absent** — V1
Biofilm and attached growth	7 mentions in passing	Thin — V1
Degrees of freedom, factorials, blocking	Part 2 chapter 17, part 19 chapter T3	Covered
Response-surface methodology as a method	Named and fitted once	**Thin** — screening, steepest ascent, design choice and canonical analysis were missing (V4)

Four chapters, V1 to V4, and eight problems, MP1 to MP4 and DE1 to DE4, taking the set to **133**.

Chapter V1 · extends weeks 44 and 48

Multiphase and solid-state fermentation

A stirred tank is two phases with a third dissolved in one of them. Add a solid and three things change at once: the transfer coefficient, the working volume, and what the reactor can be.

What "multiphase" actually means here

Phases	Example	What governs it
Liquid + gas	Ordinary submerged fermentation	k_La, mixing (part 11 ch 44)
Liquid + liquid + gas	Two-phase in situ removal	Partition coefficient, phase volume (part 3 ch 45–47)
Liquid + solid + gas	Immobilised cells, biofilm, slurry reactor	Solids holdup, intraparticle diffusion, bed hydrodynamics
Solid + gas, no free liquid	Solid-state fermentation on bran, husk, bagasse	Water activity, evaporative cooling, conduction

Three-phase reactors: the three penalties

The transfer penalty. Solids reduce the liquid fraction and damp bubble breakup, so k_La falls roughly as $(1 - \varepsilon_s)^{2.5}$ with solids holdup ε_s. At 10% solids that is a 26% loss, at **20% a 43% loss**, at 30% 59%. Since you added the solids to retain biomass, you are trading oxygen transfer for cell retention, and whether that is favourable depends on whether the process was oxygen-limited to begin with.

The volume penalty. A 20% bead loading is 20% of the vessel that is not broth — on top of the gas holdup of chapter T4. Together, 20% solids and 10% gas leave 70% of the vessel working.

The hydrodynamic constraint. Whether a bed is packed, fluidised or slurry depends on the liquid velocity against the particle's terminal velocity. For a 2 mm alginate bead with a 100 kg/m³ density difference, the terminal velocity is about **0.077 m/s**, and a 4 mm bead about 0.109 m/s. A column run at a typical 0.01 m/s superficial velocity is therefore **a packed bed, not a fluidised one** — and the difference matters, because a packed bed channels, compacts under its own weight and cannot be aerated through the bed without flooding it.

Which configuration to choose follows from those three. **Stirred tank with beads** for a small bead loading and an oxygen-demanding organism; **packed bed** for an anaerobic or microaerobic process with a clean feed; **fluidised or expanded bed** when you need both retention and mass transfer and can supply the velocity; **airlift with beads** when shear must be low, which chapter T4 says matters for the beads rather than the cells.

Biofilm and attached growth

A biofilm is the same problem with the solid self-assembled. The governing number is the **oxygen penetration depth**, $\sqrt{(2DC^*/R_V)}$, with R_V the volumetric uptake rate. For a film at 100 kg/m³ of wet biomass consuming 4 mmol $O_2 \cdot gDW^{-1} \cdot h^{-1}$:

Film density	Oxygen penetration
50 kg/m³	106 µm
100 kg/m³	**75 µm**
200 kg/m³	53 µm

So at 100 kg/m³ the active layer is 75 µm, and a film thicker than that is partly dead weight: a 300 µm film is **75% inactive** and a 1 mm film **92% inactive**. That is why biofilm reactors are designed around **thickness control** — shear to slough the excess, or a support geometry that keeps films thin — rather than around maximising attached biomass, which is the intuitive but wrong objective. The dead interior is not merely idle: it ferments, acidifies and eventually detaches as a plug.

Solid-state fermentation

With no free water the familiar levers disappear. There is no k_La, because oxygen moves through the gas phase in the inter-particle voids and then a short way into a wet particle; there is no jacket cooling worth the name, because the bed conducts badly; and pH cannot be controlled by addition.

Heat is the whole problem. At 460 kJ per mole of oxygen and 2 kg O_2 per kg of biomass, a bed growing at 3 $kg \cdot m^{-3} \cdot h^{-1}$ releases **86 $MJ \cdot m^{-3} \cdot h^{-1}$, which is 24 kW/m³** — against the 1 to 5 kW/m³ of a stirred tank. It leaves by

evaporation, which costs water: **38 kg of water per m³ per hour**, so a bed at 50% moisture holding 250 kg/m³ of water **dries out in 7 hours** at that growth rate, and in 20 hours at 1 kg·m^{-3}·h^{-1}. Hence the two controls that define SSF practice — **humidified air at high flow**, and **water replenishment** — and the reason productivity is capped by aeration and humidity rather than by oxygen transfer in the usual sense.

Conduction sets the depth. With a bed conductivity near 0.2 W·m^{-1}·K^{-1}, a heat generation of 1 kW/m³ and conduction only, the centre of a bed rises by $qL^2/2k$: **6 K at a 5 cm half-depth, 56 K at 15 cm, and 225 K at 30 cm** — the last two being thermally impossible, which is why static trays are shallow and anything deeper is forced-aerated, rotated or agitated. The reactor classes follow: trays (shallow, labour-intensive, easy), packed beds with forced aeration (deeper, with an axial temperature and moisture gradient), and rotating drums or intermittently mixed beds (uniform, with shear on the mycelium).

Solids penalty	$k_La \propto (1 - \varepsilon_s)^{2.5}$; 43% loss at 20% solids
Bed regime	compare superficial velocity with terminal velocity (0.077 m/s for a 2 mm bead)
Biofilm	active layer $\sqrt{(2DC^*/R_V)}$ = 75 µm at 100 kg/m³; a 1 mm film is 92% inactive
SSF heat	24 kW/m³ at 3 kg·m^{-3}·h^{-1}; 38 kg water/m³/h; bed dries in 7 h
SSF depth	$\Delta T = qL^2/2k$: 56 K at a 15 cm half-depth, 1 kW/m³

What to open a book for. Doran on heterogeneous reactions and particle mass transfer; a review of solid-state fermentation bioreactor design (Mitchell and colleagues) for the heat and water balances, which is the whole subject; and a biofilm-reactor review for thickness control.

Chapter V2 · extends week 48

In situ gel immobilisation as a process, not just a transport problem

Part 3 chapter 48 asked how deep oxygen gets into a bead. This chapter asks the questions that come before and after: how the bead is made, how cells get out of it, and what dissolves it.

Making the beads

The standard route is **ionotropic gelation**: a cell suspension in 1.5–3% sodium alginate is dripped or sprayed into a calcium chloride bath, where Ca^{2+} cross-links the guluronic acid blocks into an "egg-box" structure. Bead size comes from the droplet, so it is set by nozzle diameter, flow rate and whether you use vibration, coaxial air flow or an electrostatic field to break the stream. What matters downstream:

Bead radius	Volume	Surface/volume	Outer 50 µm as a share of volume
1.0 mm	4.19 µL	3,000 m^{-1}	**14%**
1.5 mm	14.1 µL	2,000 m^{-1}	10%
2.0 mm	33.5 µL	1,500 m^{-1}	7%

Smaller beads transfer better — chapter 48's Thiele modulus falls with radius — and have more surface per volume, which is the problem in the next section. The usual compromise is 1 to 2 mm, made by a vibrating nozzle at hundreds of beads a second.

Leakage: the cells get out

The failure mode nobody plans for is that **cells near the surface escape as they divide**. Treating the outer 50 µm as the escape zone, a 1 mm bead releases cells at roughly its outer-layer fraction times the growth rate: at $\mu = 0.3\ h^{-1}$ that is **0.042 g of free cells per gram of immobilised biomass per hour**, and 0.022 for a 2 mm bead. Over a day that is a substantial free-cell population, which matters three ways: the "immobilised" process is partly a free-cell process; the free cells are the ones your assays see; and in a continuous reactor they wash out, so you are continuously losing biomass.

The defences, in order of practicality: **a coating layer** (chitosan or poly-L-lysine, then a second alginate layer) which adds a diffusion barrier of its own; **larger beads**, trading transfer for retention; **growth limitation** inside the bead, which is the elegant answer — a nitrogen-limited feed lets the cells produce without dividing, so there is nothing to escape; and **accepting it** and designing the downstream for a mixed population.

What dissolves the gel

Calcium alginate is held together by an ion, so anything that takes the ion away destroys the bead. **Phosphate and citrate chelate calcium**, and both are ordinary medium components — a standard M9 or LB formulation will soften beads over days. Sodium and potassium at high concentration exchange with calcium and do the same more slowly. The fixes: formulate with the **lowest phosphate that supports growth** and feed phosphate separately if needed; include **a few millimolar of calcium in the medium** to keep the equilibrium on the right side; or cross-link with **barium or strontium**, which bind alginate more tightly, at the cost of toxicity considerations and regulatory awkwardness. Alternatives avoid the problem entirely — κ-carrageenan gelled with potassium, chitosan, or covalently cross-linked polyacrylamide and polyvinyl alcohol, each with its own toxicity and diffusivity.

When gel immobilisation actually pays

Part 3 chapter 48 computed the central trade: a retained-biomass reactor reached 0.27 $g \cdot L^{-1} \cdot h^{-1}$ against a free-cell chemostat's 0.65 at the same dilution rate, because the beads' diffusion limitation outweighed the higher biomass. So the honest default is that **immobilisation costs productivity**, and it has to be bought back by something else:

- **Operating above the washout dilution rate** — the one unambiguous win, since retained cells cannot wash out and a free-cell chemostat is hard-limited at μ_{max}.
- **Protecting cells from shear, solvent or inhibitor**, which is the argument in a two-phase process where the organic phase would otherwise contact the cells.
- **Reusing an expensive catalyst** across batches, which is the argument for immobilised *enzymes* more than cells.

- **Avoiding a downstream separation**, since the beads are removed by a screen rather than a centrifuge.

If none of those four applies, the calculation in chapter 48 says do not immobilise.

Gelation	1.5–3% alginate into $CaCl_2$; bead size from the droplet
Leakage	outer-layer fraction × μ: 0.042 g free cells per g per h for a 1 mm bead at μ 0.3
Gel killers	phosphate and citrate chelate Ca^{2+}; fix with low phosphate, added Ca, or Ba/Sr
When it pays	above washout, under shear or solvent, to reuse catalyst, or to avoid a centrifuge

What to open a book for. A review of cell immobilisation in alginate for the gelation chemistry and the stability data; part 3 chapter 48 of this module for the transport arithmetic, which this chapter assumes rather than repeats.

Chapter V3 · consolidates parts 2, 7, 8 and 11

The pathway-modification playbook: nine interventions and how to choose

The module teaches every one of these somewhere. What it never did was put them on one page with the number that tells you which to reach for.

Intervention	Use when	The number that decides it	Where
1. Overexpress a step	The step carries flux control	Flux control coefficient; gain = 1/(1 − C(1 − 1/m)), ceiling 1/(1 − C). At C = 0.1 a tenfold overexpression buys 9.9%	Part 2 ch 15; part 8 ch 30
2. Delete a competing branch	Carbon leaves through a known side reaction	Branch partition 1/(1 + r/F); at r = 1, raising F to 10 moves 67% → 91%	Part 8 ch 27; part 1 ch 6
3. Relieve feedback inhibition	The product inhibits its own first committed step	Removing feedback in the part 2 model gave +43%, against +8.3% from a tenfold overexpression	Part 2 ch 15; part 19 ch T3 of the enzymology
4. Swap the cofactor	The pathway demands the scarcer carrier	NADPH costs 0.5 C per molecule from the oxidative branch; a transhydrogenase costs ~0.33 ATP	Part 2 ch 19; part 14 ch B5
5. Push precursor supply	The pathway's precursor is a minor node	Biomass demand per gDW against pathway demand; acetyl-CoA is 3.75 mmol/gDW for biomass alone	Part 1 ch 7; part 14 ch B7
6. Engineer export	Product accumulates inside or inhibits	Intracellular ceiling 20 g/L at 50 gDCW/L; an export cap below 6 killed a coupled strain outright	Part 3 ch 45; part 11 ch 39
7. Dynamic control	The best expression differs between growth and production	Two-stage optimum: f = 0.078 of substrate to growth, 23% better than one stage	Part 8 ch 27; part 18 PP8
8. Growth coupling	Revertants will take over a long run	Guaranteed product at maximum growth; ΔPYK ΔACK gives 0.90 mol/mol at 0.072 h^{-1}	Part 11 ch 39; part 18 PP7
9. Compartmentalise or scaffold	An intermediate is toxic, volatile or shared	Peroxisomal flux 2.8 × 10^{-8} against carrier-mediated 1.7 × 10^{-7} mol·m^{-2}·s^{-1}	Part 6 ch 58; part 5 ch 53

The order to try them in

The order is not arbitrary; it follows from cost and from what the measurement tells you.

First, find out what limits it. Part 2 chapter 11's diagnosis: measure the intermediate pools. An intermediate that accumulates says the step after it is limiting; one that is absent says the step before it is. A product that rises with uptake rate says throughput, not pathway. Doing this first saves most of the work, because the common finding is that the limitation is not where the proposal assumed.

Then the cheap, high-leverage interventions: feedback relief (3) and branch deletion (2) are single point mutations or single knockouts with large effects. Feedback relief in particular gave five times the benefit of a tenfold overexpression in the part 2 model, and it costs one allele.

Then the supply-side interventions: cofactor (4) and precursor (5), which need a balance calculation first, because they trade carbon for reductant and can lower yield while raising rate.

Then export (6), which is cheap to test and catastrophic to ignore — part 11 chapter 39 showed a coupled strain made infeasible by a transport cap.

Only then expression tuning (1) and libraries, because without a control coefficient you are guessing, and the ceiling of 1/(1 − C) caps what tuning can ever deliver.

Dynamic control (7) and growth coupling (8) last, because they are architectural: they change how the process is run, not just what the strain contains, and they are worth the effort only once the pathway itself works.

Compartmentalisation (9) when a specific intermediate problem demands it — toxicity, volatility or competition for a shared pool — and not as a general improvement.

The honest caveat from part 11 chapter 39: a model proposes these, measurement decides them, and the published strains almost always contain one intervention the model could not suggest — a transporter, a regulator, or an evolved tolerance.

Chapter V4 · extends week 17 and chapter T3

Design of experiments, properly: from screening to a response surface you can defend

Chapter T3 gave the four tools. This is the method: how many runs, which design, how to walk to the optimum, and how to read the output without fooling yourself.

Degrees of freedom: the bookkeeping that decides whether a design works

Every run gives one degree of freedom. Every term you estimate spends one. What is left is the error degrees of freedom, and without it you have no test at all.

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df_error = N − 1(mean) − k(main effects) − k(k−1)/2(two-factor interactions) − …
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Design	Runs	df spent on the model	df left for error
2^2 with 4 replicates	16	4	12
2^3 with 2 replicates + 3 centre	19	7	12
2^4 unreplicated + 5 centre	21	11	10
2^4 unreplicated, no centre points	16	11	**5**
2^5 unreplicated	32	16	16

Two lessons. An unreplicated factorial with all interactions fitted leaves very little error — the 2^4 case leaves five — which is why unreplicated designs are analysed by pooling the high-order interactions into error or by a normal plot of effects instead of by F tests. And **centre points are the cheapest degrees of freedom you can buy**: three to five of them give a pure-error estimate independent of the model and let you test curvature before you commit to a quadratic.

Screening: fractional factorials and what gets confounded

With k factors a full factorial is 2^k runs, which is 128 at k = 7. A fraction assigns extra factors to interaction columns: set E = ABCD, and you get a half fraction in 16 runs. The price is **aliasing** — each effect is confounded with others, read off the defining relation.

Design	Runs	Generators	Resolution	What is confounded
2^7 full	128	—	—	Nothing
2^{7-1}	64	G = ABCDEF	VII	Nothing that matters
2^{5-1}	16	E = ABCD	**V**	Main with four-factor, two-factor with three-factor — all main effects and two-factor interactions estimable
2^{7-4}	8	E = ABC, F = BCD, G = ACD	**III**	A = BE = CF = DG — main effects confounded with two-factor interactions; screening only

The rule of thumb: **resolution III for screening many factors** when you only want to know which few matter and are willing to be fooled by an interaction; **resolution IV** when main effects must be clean of two-factor interactions; **resolution V** when you want the two-factor interactions too. Moving from 8 runs to 16 buys you from III to V at five factors, which is almost always worth it.

A worked screening, computed: five factors in a 2^{5-1}, with true effects A = 2.0, B = 1.2, C = 0.2, D = −1.5, E = 0.1 and an AB interaction of 0.7, measured with 0.4 of noise. The estimated effects (conventionally twice the coefficients) came out **A +4.11, D −2.67, B +2.32, AB +1.64**, with everything else under 0.3 — so the design correctly identifies A, B, D and the AB interaction and correctly rejects C and E. That is what screening is for: **reduce five factors to three** before spending runs on a surface.

Steepest ascent: getting to the right neighbourhood

A first-order model tells you the direction of improvement, not the optimum. The path of steepest ascent moves in proportion to the coefficients: with coded coefficients $b_A = 1.0$, $b_B = 0.6$ and $b_D = -0.75$, you move **one coded unit in A for every 0.60 in B and −0.75 in D**. Converting to real units with step sizes of 5 g/L, 0.5 °C and 100 rpm per coded unit gives a concrete path:

Step	A	B	D
1	+5 g/L	+0.30 °C	−75 rpm
3	+15 g/L	+0.90 °C	−225 rpm
5	+25 g/L	+1.50 °C	−375 rpm

You run single experiments along the path until the response stops improving, and **that is where the response-surface design goes**. Skipping this step is the commonest waste in an optimisation campaign: a central composite design centred in the wrong place fits a plane and tells you only to move, which is what the screening design already said.

Choosing the response-surface design

Factors	Quadratic coefficients	Central composite	Half-fraction CCD	Box–Behnken
2	6	13	—	—
3	10	19	—	15
4	15	29	21	27
5	21	47	31	46

The axial distance for a rotatable design is $\alpha = (2^k)^{1/4}$: **1.414 at two factors, 1.682 at three, 2.000 at four**. Rotatability means the prediction variance depends only on distance from the centre, which is what you want when you do not know the direction of the optimum. **Box–Behnken has no corner points**, so no run combines all factors at their extremes — use it when the corners are infeasible or dangerous, and accept slightly poorer estimation at the edges.

Reading the output: ANOVA, lack of fit, canonical analysis

A worked three-factor central composite design, 19 runs with five centre points, fitted to the full quadratic:

$$y = 42.01 + 3.05x_1 + 1.87x_2 - 0.43x_3 - 2.53x_1^2 - 1.85x_2^2 - 0.66x_3^2 - 1.32x_1x_2 + 0.38x_1x_3 + 0.04x_2x_3$$

Model test: $R^2 = 0.9941$, adjusted $R^2 = 0.9882$, $F = 168.0$ on 9 and 9 degrees of freedom, $p = 6 \times 10^{-9}$. Report the *adjusted* R^2, because the plain R^2 rises whenever you add a term; a large gap between the two means you have fitted noise.

Lack of fit: $F = 2.27$ on 5 and 4 degrees of freedom, $p = 0.223$ — **not significant, so the quadratic is adequate**. This is the test that most published response surfaces omit, and it is the only one that asks whether the *model form* is right rather than whether the coefficients are non-zero. A significant lack of fit with a high R^2 means the surface is real and your equation is the wrong shape for it.

Canonical analysis: write the quadratic terms as a matrix **B** and solve $x^* = -\tfrac{1}{2}B^{-1}b$ for the stationary point. Here that is **x* = (0.505, 0.323, −0.171)** with a predicted response of **43.12**, against a true optimum at (0.510, 0.313, −0.247) — recovered to within the noise. The eigenvalues of **B** are **−2.945, −1.458, −0.642**, all negative, so the stationary point is a **maximum** rather than a saddle or a minimum.

The eigenvalues say more than that. The smallest in magnitude, −0.642, with eigenvector **(0.12, −0.05, 0.99)**, identifies the **flattest direction** — a ridge along x_3 in which the response changes very little. That is operationally valuable: along a ridge you can move a factor freely to satisfy some other constraint — cost, a regulatory limit, an equipment limit — at almost no cost in response. A response surface's eigenvectors are where the process flexibility is, and almost nobody reports them.

Degrees of freedom	`df_error = N − 1 − k − k(k−1)/2 − …; centre points are the cheapest df`
Resolution	`III screening · IV clean main effects · V clean two-factor interactions`
Steepest ascent	`move in proportion to coded coefficients until the response stops improving`
Rotatable α	`(2k)1/4 = 1.414, 1.682, 2.000 at k = 2, 3, 4`
Adequacy	`adjusted R², then lack of fit against pure error from the centre points`
Canonical analysis	`x* = −½B⁻¹b; eigenvalue signs classify it; the smallest gives the ridge`

What to open a book for. Box, Hunter & Hunter for screening, aliasing and steepest ascent — chapters 5 to 12; Myers,

Montgomery & Anderson-Cook, *Response Surface Methodology*, for canonical analysis and design comparison, which is where the eigenvalue treatment above comes from.

Problems MP1–MP4 and DE1–DE4

MP1 · extends chapter V1 · 1.5 h

What do the beads cost you?

1. For solids holdup of 0, 10, 20 and 30%, compute the relative k_La from $(1 - \varepsilon_s)^{2.5}$ and the working volume lost.
2. With 10% gas holdup as well, what fraction of the vessel is broth at 20% solids?
3. Compute the terminal velocity of a 2 mm and a 4 mm bead at a 100 kg/m³ density difference.
4. At a superficial liquid velocity of 0.01 m/s, what reactor do you have, and what follows?

Worked solution. Relative k_La is 1.00, 0.74, **0.57** and 0.41 at the four loadings, so 20% solids costs **43% of oxygen transfer** and 20% of the volume. With 10% gas as well, only **70%** of the vessel is broth. Terminal velocities are **0.077 and 0.109 m/s**, both far above 0.01 m/s — so at that velocity the bed is **packed, not fluidised**. That matters because a packed bed channels, compacts under its own weight, and cannot be aerated through the bed without disrupting the flow: you must either raise the velocity by an order of magnitude to fluidise, aerate externally in a loop, or accept an anaerobic or microaerobic process.

What it teaches: adding a solid phase costs transfer and volume simultaneously, and the liquid velocity decides which reactor you actually have rather than which one you intended.

MP2 · extends chapter V1 · 1.5 h

How thick should a biofilm be?

1. Compute the oxygen penetration depth at biofilm densities of 50, 100 and 200 kg/m³, taking 4 mmol $O_2 \cdot gDW^{-1} \cdot h^{-1}$, $D = 1.5 \times 10^{-9}$ m²/s and $C^* = 0.21$ mM.
2. At 100 kg/m³, what fraction of a 50, 100, 300 and 1,000 µm film is active?
3. What is the design objective, and what are the two ways to achieve it?
4. What does the dead interior do, and why is that worse than being idle?

Worked solution. Penetration is **106, 75 and 53 µm** at the three densities. At 100 kg/m³ a 50 µm film is fully active, a 100 µm film 75%, a 300 µm film **25%** and a 1 mm film **8%** — so thicker is emphatically not better. The design objective is therefore **thickness control, not maximum attached biomass**, achieved either by shear that continuously sloughs the excess or by a support geometry that limits it.

The dead interior is worse than idle because it is anaerobic and metabolically active: it ferments, produces acids and gas, weakens the film's attachment, and eventually detaches as a plug that blocks the bed or appears in the product. A biofilm reactor that is not controlling thickness is accumulating a failure.

What it teaches: in any attached-growth system the active layer is tens of microns, so the engineering variable is thickness rather than quantity.

MP3 · extends chapter V1 · 2 h

Solid-state fermentation: the heat and water balance

1. At 2 kg O_2 per kg biomass and 460 kJ/mol O_2, compute the heat release for growth of 1, 3 and 10 $kg \cdot m^{-3} \cdot h^{-1}$, in $MJ \cdot m^{-3} \cdot h^{-1}$ and kW/m³.
2. If all of it leaves by evaporation, how much water per m³ per hour?
3. A bed at 50% moisture holds 250 kg of water per m³. How long until it dries at 1 and 3 $kg \cdot m^{-3} \cdot h^{-1}$?
4. With $k = 0.2$ $W \cdot m^{-1} \cdot K^{-1}$ and conduction only, compute the centre temperature rise at half-depths of 5, 15 and 30 cm for 1 kW/m³. What reactor designs does this force?

Worked solution. Heat is **28.8, 86.2 and 287.5 $MJ \cdot m^{-3} \cdot h^{-1}$ — that is 8.0, 24.0 and 79.9 kW/m³**, against 1 to 5 kW/m³ in a stirred tank. Evaporating all of it needs **12.7, 38.2 and 127.2 kg of water per m³ per hour**, so a bed holding 250 kg/m³ **dries out in 20 hours at the slow rate and 7 hours at the middle one**. That is the central fact of SSF: the process is limited by humidity and aeration, and water replenishment is a continuous operation rather than a formulation decision.

Conduction gives centre rises of **6 K at 5 cm, 56 K at 15 cm and 225 K at 30 cm** for only 1 kW/m³. The last two are impossible, which forces the reactor classes: **shallow trays** (passive, labour-heavy), **forced-aerated packed beds**

(deeper, with axial temperature and moisture gradients that must be managed), and **rotating drums or intermittently mixed beds** (uniform, at the cost of shear on mycelium and of mechanical complexity).

What it teaches: in solid-state fermentation the oxygen problem is replaced by a heat and water problem that is an order of magnitude harder, and bed depth is the variable it constrains.

`MP4` · extends chapter V2 · 1.5 h

Gel beads: leakage, stability and whether to bother

1. For beads of radius 1.0, 1.5 and 2.0 mm, compute the volume, the surface-to-volume ratio, and the outer 50 µm as a fraction of volume.
2. Taking that outer layer as the escape zone and $\mu = 0.3\ h^{-1}$, compute free-cell generation per gram of immobilised biomass per hour.
3. Name the two medium components that dissolve calcium alginate, and three fixes.
4. Given part 3 chapter 48's result that immobilisation cost productivity, state the four conditions under which it still pays.

Worked solution. The outer 50 µm is **14% of a 1 mm bead's volume, 10% of a 1.5 mm and 7% of a 2 mm**, and surface-to-volume falls from 3,000 to 1,500 m^{-1}. At $\mu = 0.3\ h^{-1}$ the free-cell generation is **0.042 g per g of immobilised biomass per hour for a 1 mm bead and 0.022 for a 2 mm** — a day of that is a substantial free population, so an "immobilised" process is partly a free-cell process and your assays will mostly see the escapees.

Phosphate and citrate chelate the calcium and soften the beads; ordinary M9 or LB will do it over days. Three fixes: minimise phosphate and feed it separately; include a few millimolar calcium in the medium; or cross-link with barium or strontium, accepting the toxicity and regulatory cost. The four conditions that justify immobilisation at all: **operating above the washout dilution rate**; **protecting cells from shear, solvent or inhibitor**; **reusing an expensive catalyst**; and **avoiding a centrifugation step**. Absent all four, chapter 48's productivity comparison says do not.

What it teaches: immobilisation is a retention technique with three running costs — diffusion, leakage and gel stability — and it needs a specific reason, not a general one.

`DE1` · extends chapter V4 · 1 h

Degrees of freedom: will this design even work?

1. For a 2^2, 2^3, 2^4 and 2^5 design with the replication and centre points stated in chapter V4, compute the degrees of freedom spent and left.
2. Which design leaves too few, and what are the two standard remedies?
3. How many centre points would you add to a 2^4, and what do they buy?
4. Why can an unreplicated factorial still be analysed?

Worked solution. Spent and left: 2^2 with four replicates, 4 and **12**; 2^3 with two replicates and three centre points, 7 and **12**; 2^4 unreplicated with five centre points, 11 and **10**; 2^4 unreplicated with none, 11 and **5**; 2^5 unreplicated, 16 and **16**.

The bare 2^4 is the weak one at five error degrees of freedom. Two remedies: **pool the high-order interactions into error**, on the argument that three- and four-factor interactions are rarely real, which recovers degrees of freedom at the cost of an assumption; or **add centre points**, which give pure error independent of the model. I would add **four to five centre points** to a 2^4 — they cost 25% more runs and buy an independent error estimate plus a curvature test that tells you whether to go quadratic at all. An unreplicated factorial is still analysable without any error estimate by a **normal plot of the effects**: real effects fall off the line that the inactive ones define.

What it teaches: count the degrees of freedom before running anything, because a design with none left cannot be tested, only described.

`DE2` · extends chapter V4 · 1.5 h

Screening five factors, and knowing what you confounded

1. Build a 2^{5-1} with E = ABCD and write the defining relation and the alias of A and of AB.
2. From the simulated data in chapter V4, rank the estimated effects and decide which are real.
3. Compare with a 2^{7-4} in eight runs: what is confounded, and when is that acceptable?
4. How many runs for a full 2^7, a half fraction, and the eighth fraction?

Worked solution. With E = ABCD the defining relation is I = ABCDE, so A = BCDE and AB = CDE: every main effect is

aliased with a four-factor interaction and every two-factor interaction with a three-factor one — **resolution V**, and for practical purposes everything of interest is clean. The estimated effects rank **A +4.11, D −2.67, B +2.32, AB +1.64**, then nothing above 0.29 — so A, B, D and the AB interaction are real and C and E are noise, which matches the truth used to generate the data.

A 2^{7-4} in eight runs has A = BE = CF = DG: **main effects confounded with two-factor interactions**, resolution III. That is acceptable only for a first screen where you expect a few large main effects and will follow up — and it is dangerous when interactions are likely, because a real AB can appear as a main effect of a factor that does nothing. Run counts: **128 full, 64 half, 8 for the eighth fraction**, which is the trade in one line.

What it teaches: a fractional design is defined by what it confounds, and you must be able to write the aliases before you trust the effects.

`DE3` · extends chapter V4 · 1 h

Walk the path of steepest ascent

1. Given coded first-order coefficients of 1.0 for A, 0.6 for B and −0.75 for D, write the direction of steepest ascent.
2. Convert to real units with step sizes of 5 g/L, 0.5 °C and 100 rpm per coded unit, for steps 1 to 5.
3. How do you know when to stop walking?
4. What happens if you skip this step and place a central composite design at the screening centre?

Worked solution. The direction is proportional to the coefficients, so normalising on the largest: **one coded unit in A for every 0.60 in B and −0.75 in D**. In real units that is +5 g/L, +0.30 °C and −75 rpm per step, reaching +25 g/L, +1.50 °C and −375 rpm at step five. You run **single, unreplicated experiments along the path** and stop when the response stops improving — two consecutive non-improving points is the usual rule — and that point is where the response-surface design goes.

Skipping it is the commonest waste in an optimisation campaign. A central composite design centred far from the optimum fits an almost-planar surface: the quadratic terms come out small and insignificant, the stationary point lands outside the design region where the model has no support, and the whole 19-run design tells you only to move in a direction the 16-run screen had already given you.

What it teaches: steepest ascent is cheap, unreplicated and the step that makes a response-surface design worth running.

`DE4` · extends chapter V4 · 2 h

Read a response surface: adequacy, optimum, and the ridge

1. From the fitted equation in chapter V4, report R^2, adjusted R^2 and the model F test, and say which you would quote.
2. Carry out the lack-of-fit test and state what a significant result would have meant.
3. Solve for the stationary point and classify it from the eigenvalues.
4. Identify the flattest direction and say what you would do with it.

Worked solution. $R^2 = 0.9941$, adjusted R^2 = **0.9882**, F = 168.0 on 9 and 9 df, $p = 6 \times 10^{-9}$. Quote the **adjusted** R^2, because plain R^2 rises with every term added; a large gap between them is the signature of fitted noise. Lack of fit gives **F = 2.27, p = 0.223** — not significant, so the quadratic form is adequate. A *significant* lack of fit with a high R^2 would have meant the surface is real and the equation is the wrong shape for it, and the response would be to add higher-order terms, transform a variable, or shrink the design region.

The stationary point is **x* = (0.505, 0.323, −0.171)**, predicted response **43.12**, against a true optimum of (0.510, 0.313, −0.247) — recovered within the noise. Eigenvalues are **−2.945, −1.458, −0.642**, all negative, so it is a **maximum**.

The flattest direction is the eigenvector of the smallest-magnitude eigenvalue, **(0.12, −0.05, 0.99)** — essentially x_3. Along that ridge the response barely changes, so **x_3 can be set by something other than the response**: cost, an equipment limit, a regulatory constraint, or operator convenience. Reporting the ridge is how a response surface becomes operationally useful rather than a single set-point, and almost no published surface does it.

What it teaches: the optimum is the least interesting output of a response surface; the adequacy test and the flat directions are what you actually use.

Closing notes to part 20

Direct answers to what you asked

Multiphase fermentation — the two-liquid-phase and gas-fermentation cases were covered (part 3 chapters 45 to 47, part 5 chapter 55). Three-phase gas–liquid–solid and solid-state were **not**, and biofilms were a passing mention. Chapter V1 and problems MP1 to MP3 cover them now.

In situ gel fermentation — the transport side was covered properly in part 3 chapter 48, and the *process* side was not: bead production, cell leakage, gel dissolution by phosphate, and the four conditions under which immobilisation pays. Chapter V2 and problem MP4.

Metabolic pathway modification — covered thoroughly but scattered over twelve parts. Chapter V3 is the consolidation: nine interventions, the number that decides each, where it is taught, and the order to try them in.

Design of experiments and response-surface methodology — degrees of freedom, factorials and blocking were covered; the *method* was not. Chapter V4 now runs screening → aliasing → steepest ascent → design choice → ANOVA → lack of fit → canonical analysis, with every number computed, and problems DE1 to DE4 work each stage.

The numbers worth keeping

- 20% solids loading costs **43% of k_La** and 20% of the working volume.
- A 2 mm bead's terminal velocity is **0.077 m/s**, so a 0.01 m/s column is a packed bed.
- A biofilm's active layer is **75 µm** at 100 kg/m³; a 1 mm film is **92% inactive**.
- Solid-state fermentation at 3 kg·m^{-3}·h^{-1} releases **24 kW/m³** and dries its bed in **7 hours**.
- A 15 cm half-depth bed at 1 kW/m³ runs **56 K** hot at the centre.
- A 1 mm alginate bead leaks **0.042 g of free cells per gram per hour** at $\mu = 0.3$.
- Feedback relief gave **+43%** where a tenfold overexpression gave +8.3%.
- A rotatable CCD's axial distance is **1.414, 1.682, 2.000** at two, three and four factors.
- The fitted surface recovered the optimum to **(0.505, 0.323, −0.171)** against a true (0.510, 0.313, −0.247), with lack of fit $p = 0.223$.

On the numbers

All computed here. The assumptions to challenge: the $(1 - \varepsilon_s)^{2.5}$ solids correlation, which is a rough empirical form and varies with bead density and sparger; the 50 µm escape zone in the leakage model, which is a plausible figure rather than a measurement; the bed conductivity of 0.2 W·m^{-1}·K^{-1} and the 2 kg O_2 per kg of biomass in the SSF balance; and the biofilm's wet-biomass density. The *forms* of all these relations are standard; the coefficients are yours to measure, and they belong on sheet 10 of the lookup workbook.

Where the set stands

Source	Problems
Curriculum appendix A1–W3	54
Parts 15–19 (X, Y, Z, MC, R, PP, PR)	69
Part 20 (MP1–MP4, DE1–DE4)	8
Total	**131**, with G2 and the two construction exercises making 133 in the tracker